

## 1.14

$P(AB)$  is the probability of the intersection of the event sets  $A$  and  $B$ .

Thus the intersection of the two sets is at most the size of the smaller set. Thus  $P(AB)$  is at most  $P(A)$

To find the minimum probability, its trying to minimize the intersection of sets  $A$  and  $B$ . In otherwords, maximumizing the intersection of  $A$  and  $B^c$ .  $P(A) = 0.4$  and  $P(B^c) = 1 - P(B) = 1 - 0.7 = 0.3$  The maximum of this intersection is 0.3 because  $P(B^c)$  is the smaller one, thus the remaining 0.1 probability must be in the intersection of  $A$  and  $(B^c)^c = B$ . Thus the minimum of  $P(AB)$  is 0.1

Therefore  $0.1 \leq P(AB) \leq 0.4$

## 1.18

$$\Omega = \{3, 4, 5\}$$

| $k$                  | 3              | 4             | 5              |
|----------------------|----------------|---------------|----------------|
| $p_X(k) = P(X == k)$ | $\frac{3}{16}$ | $\frac{1}{2}$ | $\frac{5}{16}$ |

## 1.26

$$|\Omega| = \binom{15}{4} = 1365$$

$$|2 \text{ men } 2 \text{ women}| = \binom{10}{2} * \binom{5}{2} = 45 * 10 = 450$$

$$P(2 \text{ men } 2 \text{ women}) = \frac{|2\text{m}2\text{w}|}{|\Omega|} = \frac{450}{1365} = \frac{30}{91}$$

## 1.28

**a**

$A$  is w/o replacement

$$P(A) = \frac{n^2 - n + m^2 - m}{(n+m)(n+m-1)}$$

$$\text{Alternatively, } P(A) = 1 - \frac{2*n*m}{(n+m)(n+m-1)}$$

**b**

$B$  is w/ replacement

$$P(B) = \frac{n^2+m^2}{(n+m)^2}$$

$$\text{Alternatively, } P(B) = 1 - \frac{2nm}{(n+m)^2}$$

**c**

Unless  $P(B) = 0$  or  $P(B) = 1$  then  $P(B) > P(A)$ . If  $P(B) = 0$  then  $m, n < 2$  so  $P(A) = 0$  and thus  $P(B) = P(A)$ . For the case of  $P(B) = 1$  then  $m$  or  $n$  is at least 2 and the other must be 0, thus  $P(A)$  is also 1 and  $P(B) = P(A)$ .

For any other case where  $m$  or  $n$  is greater than 1 and the other is non zero,  $P(B) > P(A)$

Intuitively, when you do not replace the ball, the probability you get the same color again is lower because there is less of the same color now. Therefore  $P(A)$  is smaller than  $P(B)$ .

For a more formal proof, looking at the compliments of  $A$  and  $B$ ,  $P(A) = 1 - \frac{2nm}{(n+m)(n+m-1)}$  and  $P(B) = 1 - \frac{2nm}{(n+m)^2}$ .

It is clear that  $(n+m)^2 > (n+m)(n+m-1)$  [when  $n+m > 1$ ] so that means that  $\frac{2nm}{(n+m)^2} < \frac{2nm}{(n+m)(n+m-1)}$

$$\rightarrow -\frac{2nm}{(n+m)^2} > -\frac{2nm}{(n+m)(n+m-1)}$$

$$\rightarrow 1 - \frac{2nm}{(n+m)^2} > 1 - \frac{2nm}{(n+m)(n+m-1)}$$

$$\rightarrow P(B) > P(A)$$

## 1.33

$$|\Omega| = 6^5$$

$$|FH| = 6 * 5 \binom{5}{3} = 300$$

$$P(FH) = \frac{300}{6^5} = \frac{25}{648}$$

## 1.41

$A_i = \{\text{player } i \text{ wins no games}\}$

$$P(A_i) = \frac{2^4}{3} = \frac{16}{81}$$

$A_i \cap A_j = \{\text{players } i, j \text{ win no games}\}$

$$P(A_i \cap A_j) = \frac{1}{3^4} = \frac{1}{81}$$

$A_1 \cap A_2 \cap A_3 = \{\text{no one wins any games}\}$

$P(A_1 \cap A_2 \cap A_3) = 0$  because at least one person must win each round.

$$P(A_1 \cup A_2 \cup A_3) = \sum_{k=1}^3 (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} P(A_{i_1} \cap \dots \cap A_{i_k}) = 3P(A_i) + 3P(A_i \cap A_j) + P(A_1 \cap A_2 \cap A_3) = \frac{16}{27} + \frac{1}{27} + 0 = \frac{17}{27}$$

## 1.44 (tentative)

**a**

$$X, Y \in \{1, 2, 3, 4, 5, 6\}$$

**b and c**

| $k$           | 1               | 2              | 3              | 4              | 5               | 6               |
|---------------|-----------------|----------------|----------------|----------------|-----------------|-----------------|
| $P(X = k)$    | $\frac{1}{36}$  | $\frac{1}{12}$ | $\frac{5}{36}$ | $\frac{7}{36}$ | $\frac{9}{36}$  | $\frac{11}{36}$ |
| $P(X \leq k)$ | $\frac{1}{36}$  | $\frac{1}{9}$  | $\frac{4}{9}$  | $\frac{5}{9}$  | $\frac{2}{3}$   | $\frac{1}{3}$   |
| $P(Y = k)$    | $\frac{11}{36}$ | $\frac{9}{36}$ | $\frac{7}{36}$ | $\frac{5}{36}$ | $\frac{3}{36}$  | $\frac{1}{36}$  |
| $P(Y \leq k)$ | $\frac{11}{36}$ | $\frac{5}{9}$  | $\frac{3}{4}$  | $\frac{8}{9}$  | $\frac{35}{36}$ | $1$             |